jacecarrdev@gmail.com

jacecarr.dev | GitHub | LinkedIn

JASON CARR

PROCESS DEVELOPMENT & MATERIALS R&D

PROFESSIONAL SUMMARY

Experimental R&D engineer and materials researcher with 5 years of hands-on laboratory experience developing chemical and crystal-growth processes, synthesizing and characterizing functional materials, designing and commissioning laboratory equipment, and applying statistical analysis to experimental systems. Experienced in mixing, DOE-driven process optimization, spectroscopy and microscopy, technical documentation, chemical safety, and multidisciplinary collaboration with academic, commercial, and government partners. Currently pursuing an M.S. in Electrical and Computer Engineering with additional depth in controls, instrumentation, and data analysis.

EMPLOYMENT HISTORY

JUN 2026 - PRESENT

Service Technician, Fine Fabric Care | Butte, MT

- Performs inspection, cleaning, restoration, chemical testing, and mechanical/HVAC system maintenance under IICRC standards; troubleshoots field conditions, explains processes to customers, and works independently or in supervisory roles while improving job completion efficiency.

JAN 2017 - AUG 2021

Laboratory Researcher, Cone-Thiel Spectroscopy Laboratory, Montana State University | Bozeman, MT

- Developed and optimized laboratory-scale materials processes using co-precipitation, hydrothermal synthesis, solution and flux growth, vapor transport, proton exchange, thermal deposition, and plasma deposition; controlled composition, mixing, titration, agitation, temperature, atmosphere, drying, calcination, and crystallization conditions.
- Prepared and processed solutions, powders, slurries, wafers, and crystalline materials using magnetic/mechanical mixing, sonication, dissolution, precipitation, powder blending, centrifugation, filtration, and controlled thermal treatment; calculated stoichiometry, molarity, precursor ratios, dopant concentrations, yields, and batch quantities.
- Designed and executed multivariable experimental studies for crystal growth, wafer doping, and proton-exchange processes, systematically varying chemical, thermal, temporal, and compositional parameters to isolate variables, optimize outcomes, and improve reproducibility and procedural efficiency.
- Synthesized Yb:YAG, Yb:YLiF, Eu:CaCO3, rare-earth-doped YVO4, NaCl, and LiNbO3 materials; characterized process outcomes using absorption/fluorescence spectroscopy, SEM, XRD, and AFM to relate process conditions to morphology, phase, purity, and optical response.
- Reduced, statistically analyzed, and visualized experimental datasets using Python, MATLAB, R, OriginLab, and Excel, including regression/curve fitting, uncertainty and covariance analysis, frequency-domain analysis, filtering, calibration, and scientific visualization.
- Designed, constructed, commissioned, and upgraded laboratory process equipment including a vertical Bridgman crystal-growth apparatus, thermal and plasma deposition systems, and hydrogen/oxygen electrolysis generators for controlled annealing atmospheres, expanding process capacity and throughput while improving controllability, repeatability, and laboratory-scale process scalability.
- Built and maintained an in-house experimental database linking sample identities, compositions, processing histories, experimental parameters, and characterization results to provide traceable records and support reproducibility across research programs.
- Developed test plans, SOPs, and technical reports; prepared weekly experimental reports and presented findings, conclusions, and recommended next steps at research meetings; authored grant proposals, conference posters, and technical presentations. Managed SDS documentation, chemical inventory, labeling, compatibility-based storage, hazardous-material handling/disposal, equipment calibration, and laboratory housekeeping.
- Collaborated with Boeing, the U.S. Department of Defense, AdvR, academic groups, and interdisciplinary researchers on optoelectronic crystalline signal-processing and quantum-information R&D spanning materials, optics, electronics, mechanical design, and experimental analysis; communicated technical findings across external partner teams.

NOV 2021 - FEB 2023

Team Lead, NocturnalGG

- Led a competitive technical team, analyzed performance data, coordinated schedules and external stakeholders, and presented strategic recommendations; improved average team placing by more than 400%.

MAR 2023 - PRESENT

Additional Professional Experience

- Butte Tango outreach/web analytics; T-Mobile technical sales and support (#2 Mountain West sales metrics, 98th-percentile customer satisfaction); UPS seasonal route operations with nearly 100% delivery success.

EDUCATION & TECHNICAL SKILLS

MAY 2024 - CURRENT

Master of Science, University of Colorado Boulder

Electrical and Computer Engineering (In Progress)

AUG 2015 - MAR 2019

Bachelor of Science, Montana State University

Professional Physics | Minor in Mathematics

Research emphasis: materials synthesis, crystal growth, solid-state spectroscopy, and experimental physics

PROCESS / LAB

Mixing & Material Handling · Mass & Stoichiometric Balances · Thermal Processing & Heat Transfer · Design of Experiments (DOE) · Process Optimization · SOPs/Test Plans · SDS/HSE

CHARACTERIZATION

Absorption/Fluorescence Spectroscopy · SEM · XRD · AFM · Imaging

DATA / SOFTWARE

Python · MATLAB · R · OriginLab · Excel · Data Reduction & Visualization

ENGINEERING

Equipment Troubleshooting · Instrumentation · Controls · CAD · Machining/Fabrication · Additive Manufacturing · Equipment Commissioning